

of today's games, objects just don't seem to act the way you'd want or expect them to in real life. Currently, most of the action is limited to pre-scripted or 'canned' animations triggered by in-game events like a gunshot striking a wall. Even the most powerful weapons can leave little more than a smudge on the thinnest of walls; and every opponent you take out, falls in the same pre-determined fashion. Players are left with a game that looks fine, but is missing the sense of realism necessary to make the experience truly immersive.

With NVIDIA PhysX technology, walls can be torn down, glass can be shattered, trees bend in the wind, and water flows with body and force. NVIDIA GeForce GPUs with PhysX deliver the computing horsepower necessary to enable true, advanced physics in the next generation of game titles making canned animation effects a thing of the past.

PHYSICS PROCESSING IS BEST UTILISED BY A GPU

The multithreaded PhysX engine was designed specifically for hardware acceleration in massively parallel environments. GPUs are the natural place to compute physics calculations because, like graphics, physics processing is driven by thousands of parallel computations. Today, NVIDIA's GPUs, have as many as 512 cores (GTX 580) or 1024 cores (GTX 590), so they are well-suited to take advantage of PhysX software.

CARDS SUPPORTING PHYSX

The minimum requirement to support GPU-accelerated PhysX is a GeForce 8-series or later GPU with a minimum of 32 cores and a minimum of 256MB dedicated graphics memory. However, each PhysX application has its own GPU and memory recommendations. In general, 512MB of graphics memory is recommended unless you have a GPU that is dedicated to PhysX.

PHYSX ON MULTI-GPU SETUPS

When two, three, or four matched GPUs are working in SLI, PhysX runs on one GPU, while graphics rendering runs on all GPUs. The NVIDIA drivers optimize the available resources across all GPUs to balance PhysX



<http://goo.gl/5gXGW>:

Do check out the NVIDIA GeForce GTX PhysX PC Trailer video to get a visual feel of how PhysX enriches the realism in a game with objects interacting with their surroundings. For instance from timestamp 0.34 onwards the screen splits into two showing you the gameplay in non-PhysX mode and PhysX mode.

GAME TITLE	PHYSX EFFECTS
Alice: Madness Returns	Debris and particle simulation, destructible environments, fluid simulation
7554	Rigid body simulation, vehicle simulation, destructible environments, cloth simulation
Batman: Arkham City	dynamic fog and steam, dynamic bank notes, posters etc., additional rigid bodies, destructible environments, cloth simulation, debris and particle effects
Batman: Arkham Asylum	dynamic fog and steam, dynamic tearable cloth and cobwebs, spark effects, dynamic paper and leaves, additional rigid bodies, destructible environments
Mafia II	additional debris (up to 10,000 pieces at a time), dynamic cloth, dynamic smoke
Metro 2033	debris simulation, dynamic smoke

computation and graphics rendering. Therefore users can expect much higher frame rates and a better overall experience with SLI.

A new configuration that's now possible with PhysX is two non-matched (heterogeneous) GPUs. In this configuration, one GPU renders graphics (typically the more powerful GPU) while the second GPU is completely dedicated to PhysX. By offloading PhysX to a dedicated GPU, users will experience smoother gaming.

Finally we can put the above two configurations all into one PC! This would be SLI plus a dedicated PhysX GPU. Similarly to the 2 heterogeneous GPU case, graphics rendering takes place in the GPUs now connected in SLI while the non-matched GPU is dedicated to PhysX computation.

PhysX technology is supported by gaming engines such as Unreal Engine 3, Gamebryo, Vision, Instinct, Trinigy, Diesel, among others and some of the popular game publishers who have licensed PhysX for their global studios include EA, THQ, 2K Games, Sega, among others. 



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